Mini Project 11 — Electricity Tariff and Billing Dispute Triage: Project Plan

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1. Problem and scope

A multi-agent service will investigate Karnataka billing disputes and recommend whether the bill should stand, a revision should be considered, or a site inspection is needed. Every recommendation must cite supporting provisions. A grievance officer decides; the service never changes a bill.

2. Agent design

Agent	Responsibility	Hand-off
Intake	Validate context; classify the dispute; extract premises use and load.	Structured facts to research
Research	Retrieve the apparent rule, search for modifiers and re-query.	Draft + retrieved evidence
Verifier	Check citations and support; request another search or escalate.	Verified route or human review

Tools: corpus search, clause lookup and request-field lookup, chosen by the agent. **Limits:** eight research turns, ten tool calls and 90 seconds. Unverified runs escalate.

3. Architecture

One **docker compose up**; asynchronous runs return an ID for status polling.

Service	Role / deployment reason
FastAPI + web client	Validation, quarantine, submission and status; separate from execution.
Worker + verifier	LangGraph orchestration; independent verification; scale workers.
Redis + Postgres	Queue, cache and rate limits; persistent results, feedback and traces.
Retriever + index job	FAISS/BM25 search; ingestion and re-embedding independent of agents.
Monitoring + LLM stub	Prometheus, Grafana and Jaeger; delayed stub for provider-free load tests.

4. Corpus and retrieval

Use the **KERC Combined Tariff Order 2025** and **Karnataka Conditions of Supply** with the first two amendments. LangChain retrieval combines FAISS and BM25. Clause-level chunks retain document, page, clause ID and amendment status. Local embeddings and an idempotent CLI support repeatable indexing and re-embedding.

5. The 4.1 case: reference answer

For an occupant tailoring in one room of a home, retrieve **LT-1 (printed p. 534)**, then the competing **LT-5 tailoring-shop provision (p. 539)**. On the stated facts, recommend **bill-stands-with-reason** under LT-1, citing both. Unclear occupancy or a separate establishment requires human review.

6. Evaluation, LLMOps and operations

Evaluation: 25 scenarios; compare two prompts and modifier search on/off using groundedness, context relevance, faithfulness and hallucination rate.
Safety and reliability: injection/output guards, timeouts, backoff, circuit breaker, caching and rate limits.
Evidence: trace each run; record step-level tokens and cost per 100 runs. Use Locust with the stub to test p95 ≤ 60 seconds at 0.5 runs/second and compare one versus three workers.

7. Priorities

1. Working demo: Compose, API and web client. **2. Correctness:** tailoring case, verification, quarantine and escalation. **3. Evaluation:** prompt/retrieval comparisons. **4. Operations:** traces, load tests, costs, runbook and final report.